## Q1.

Railway lines can be joined together by pouring molten iron into the gap between them.

(a) The molten iron is produced by the reaction between powdered aluminium and iron oxide. Complete the word equation for the reaction.

aluminium + iron oxide → iron + …….

1 mark

(b) Iron can be produced from a mixture of aluminium and iron oxide but **not** from a mixture of copper and iron oxide.
Write the names of the **three** metals, in the order of their reactivity.

most reactive …….

……..

……..

1 mark

(c) The list shows the names and symbols of five metals in order of their reactivity.

| **name** | **symbol** |
|---|---|
| sodium | Na |
| calcium | Ca |
| magnesium | Mg |
| zinc | Zn |
| silver | Ag |

(i) What, if anything, would be the result of heating zinc powder with calcium oxide?

……..…….

1 mark

(ii) Write down the **name** of a metal in the list that will **not** react with a solution of magnesium sulphate.

…….

1 mark

(d) The powdered metal with the symbol Zn burns in air.
Write the **word equation** for the reaction.

…….

2 marks
Maximum 6 marks

## Q1.

(a) aluminium oxide

**1 (L7)**

(b) aluminium
iron
copper

*answers must be in the correct order*
*do **not** accept 'iron oxide'*

**1 (L6)**

(c) (i) no reaction

*accept 'nothing'*
*accept 'zinc and calcium oxide'*

**1 (L7)**

(ii) any **one** from

- zinc
  *accept 'Zn'*
- silver
  *accept 'Ag'*
- magnesium
  *accept 'Mg'*

**1 (L7)**

(d) zinc + oxygen →

**1 (L7)**

zinc oxide

**1 (L7)**

**[6]**